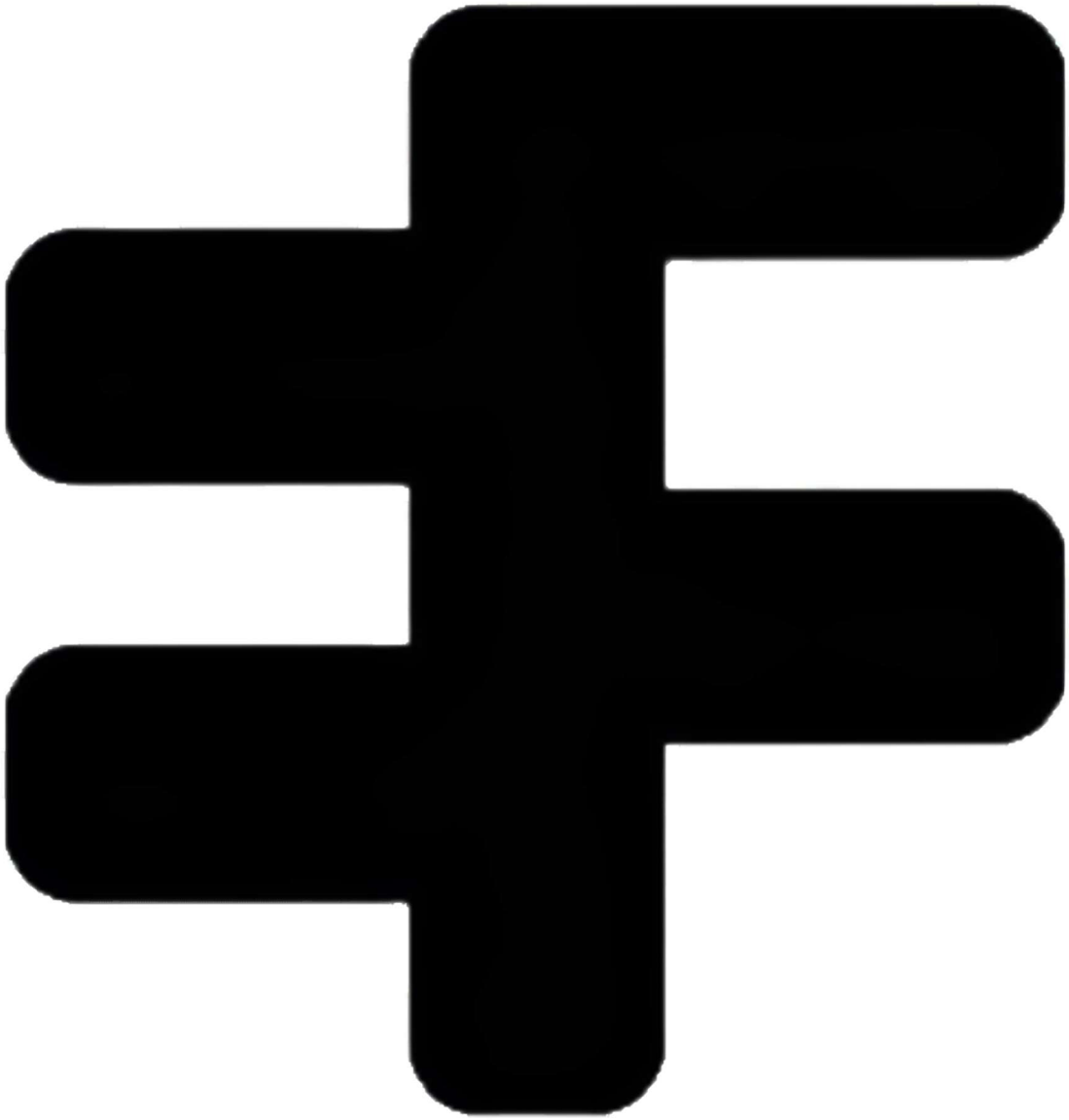

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Q : What are X-rays, how are they used, and what is their purpose?
A :Bragg's (1912) studied the behavior of X-rays when shined on crystals. Atoms in crystals cause a beam of incident X-rays to diffract into many specific directions. In the resulting image, areas with many atoms appear darker, and areas with fewer atoms appear lighter. This method is used in scientific research to analyze the internal structure of objects.

Q : Explain X-Ray diffraction and its significance.
A : It is a technique to understand internal structure of cyrstals. Initially, high energy X-rays are sent to a crystal, and the emitted rays form a pattern on a screen. It provides details about the crystal structure. Bragg's studied this behavior and determined a mathematical relationship between the wavelength of the X-Rays and the distance between the atoms in a crystal through reflection.

Q : What is Stereochemistry? How do computer scientists work on this? One of the areas where computer scientist work is the prediction of 3-D structure of molecules. What do they consider in these estimations.

A :Stereochemistry explores the three-dimensional (3D) arrangement of molecular structures in chemistry. chemical compunds. Computer scientists working on stereochemistry use computer models to simulate the geometry and graphics of molecular structures. This plays a crucial role in drug design and molecular interaction analysis.

Q : Explain periodic table and significance ? TARİHÇE

A : Mendelev started up the table of elements in order of their atomic weight, every so often their properties started to repeat. He made a table of repeating periods. When an element did not seem to be similar in properties to the are above it, he skipped a position.Then, Henry Mosaley examined the elements with an x-ray spectroscope and observed that they were arranged in a staircase-like pattern. Based on his experiments, he realized that the periodic table should be organized not according to atomic mass but according to protons (nuclear charge). All discrepancies in the periodic table were resolved.

Q : When the Curies entered the dark lab one day, Pierre and Marie noticed that the radium glowed like a light. How could a matter, which had no interaction to take energy from the environment, glow ?
A : They realized that radioactivity was atoms falling apart and spitting out energy. That meant there must be smaller pieces inside, still awaiting discovery.Radium could emit light and release energy due to its spontaneous radioactive decay process, even without interacting with the environment.

Q: Allotrope

Allotropes are different crystal structures of the same element. Lavoisier showed that burning diamonds and graphite produced the same gas.
Q : What do we do to improve science?

A : Besides sharing observations and experiments to advance science, the terminology is

of great importance. Nomenclature system provides clear and precise scientific communication. A good naming system makes sharing complex information easy.

Q : How did Glenn Seaborg's discovery of plutonium opened up a whole new field of elements that are still being studied today.

A: Seaborg bombarded uranium with neutrons and found plutonium. These neutrons that hit uranium turn into protons and it becomes a new element.
This expanded the Periodic Table and provide synthesis and study of elements with atomic numbers greater than that of uranium. The study of these elements, including plutonium, has played a critical role in nuclear science and technology. Researchers continue to explore their properties and applications today. Seaborg's work significantly impacted our understanding of elements and their practical uses.

Q : What is produced from uranium?

A : Uranium is used to generate nuclear energy through fission reactions and produce various radioactive materials.

Q : How Henry Moseley's atomic number research redefined the Periodic Table.

A : Harry Moseley's research on atomic numbers redefined the Periodic Table by organizing elements based on their atomic numbers rather than atomic weights. Now, each element is arranged according to the number of protons in its nucleus. Moseley's work contributed to a more meaningful and accurate structure for the table.

Q : :Same chemical composition but different physical features of compounds are due to differences in what?

A : These compounds named isomers.Compounds with the same chemical composition but different physical properties usually result from differences in molecular arrangement or structure.

Q :Atom has 3 parts. Which part would you use to hit an atom (for instance uranium) and Why ?

A : To hit atoms, you usually target their nuclei. Energy is needed, and particles like neutrons are used. Especially with heavy elements, this bombardment can release energy.

Q : Studying gases helped the development of chemistry, how ?

A : The gases allowed measurement of weight, pressure, volume and temperature. It was quite sensitive, easy to carry and measure, and influence other things less. This situation led the foundation for modern chemistry.

Q : How does day to day differences change the height of the Mercury in the barometer?

A : If the height and temperature are changed, the height of mercury will change. It means the balance of pressure for gases.

Q : Why do you think the development of analytical balance is significant ?

A : It far exceeded the accuracy of any other balance of the time and became an important scientific instrument in most chemistry laboratories

Q :What is called “Phlogisticated Air” and “Dephlogisticated Air” and why ?

A : The mixture of nitrogen and carbon dioxide was called Phlogisticated Air. On the other hand, when oxygen was first discovered, it was called Dephlogisticated Air because it had a large affinity for phlogiston, it must be particularly devoid of it. DEPHLOGISTICATED Air supports the life and the PHLOGISTICATED Air does not.

Q : How did Lavoisier discovered H and O ?

A : Lavoisier observed that there were quantitative reactions involving heat. Heat added to water gave steam discovered that air reacting to make calx lost a bit. He thought air contains an element. Joseph Priestley discovered a new gas. Lavoisier conducted his own experiments and announced the discovery of a new and pure air, which he named oxygen considering Priestley's findings. This discovery was beginning of Lavoisier's groundbreaking contributions to modern chemistry and leading to the end of the phlogiston theory. Lavoisier named cavedish's phlogiston hydrogen

Q : Explain Dalton's particle chemistry ?

A : - All matter is made of atoms. Atoms are indivisible and indestructible.
- All atoms of a given element are identical in mass and properties.
- Compounds are formed by a combination of two or more different kinds of atoms.
- A chemical reaction is a rearrangement of atoms.

Q : Explain Law of Multiple Proportions and how is it related to Law of Partial Pressures ?

A : The law of multiple proportions says that when two elements combine with each other to form two or more compounds, the ratios of the masses of one element that combines with the fixed mass of the other are simple whole numbers. Law of Partial Pressures says that the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture. -- PV α n --

Q : What was the basis of all things up until the 18th century?

A : Earth, water, air and fire was the basis of all things. There wasn't scientific basis until 18th century and not a theory for chemistry.

Q : What is the conservation of mass in a reaction, why is it significant in the development of chemistry ?

A : In a chemical reaction, matter is neither created nor destroyed. According to the law of conservation of mass, the mass of the products in a chemical reaction must equal the mass of the reactants. This principle supports stoichiometry, helps estimate quantities, and is important for balancing chemical equations and validating experimental results. It is essentially the fundamental of modern chemistry.

Q : What is spectroscope used for?

A : Various elements were known to give discrete spectra when heated. By placing the element in a fire, the light produced, when broken up by a prism, consisted of only a few spectra lines instead of the continuum of a rainbow. Spectra lines were fingerprints of the elements due to crystalline. Spectroscope is the device that analyze this spectrum.

Q : Why do we need a spark of electricity for the combustion of Hydrogen and Oxgyen?

A : Hydrogen and oxygen react in a chemical formula to synthesise water. The spark is the activation energy for begining his reaction. Breaking bonds takes in energy. Creating bonds releases energy.

Q : What is the principle of Eudiometer ?

A : Aplications of a eudiometer include the analysis of gases and determination of volume differences in chemical reactions. The eudiometer is filled with water, inverted so that its

open end is facing the ground and then submersed in a basic of water. A chemical reaction is taking place through which gas is created.

Q :How did atoms combine to make different compounds ?

A : There is two types of bounding. One of them is electron transfer, ionic bond. The other one is electron sharing, covalent bond. If elements share electrons equal, then its non polar covalent. If one of them pull more, it causes dipole and it is polar covalent. (efe)

Q :What is chemical algebra, describe with an example ?

A : Chemical algebra is also known as Stoichiometry. It is founded on the law of conservation of mass where the total mass of the reactants equals the total mass of products.

Q: Different chemical compounds but same composition of elements. How do you explain ?

A :When the elements are joined, the atoms lose their individual properties and have different properties from the elements they are composed of. A chemical formula is used a quick way to show the composition of compounds. Letters, numbers, and symbols are used to represent elements and the number of elements in each compound.

Q :How did Anton van Leeuwenhoek accelerate biological studies?

A : He invented improved lenses. Then the compund lenses magnified object 500 times. Later the electron microscope marvelously advanced biological research. It is 20 times as sensitive as the light microscope.

Q : How did the discover of Robert Hooke help the development of biology ?

A : Cells are discovered by Robert Hooke and he investigate cells in 17th century. He also invented the microscope.

Q : What is miasma theory ?

A : The miasma theory states that diseases and sicknesses come from breathing in a miasma. It would effect people where the area that have such vapors.

Q : Why is Griffith's experiment important? Please explain the experiment.

A : The first experiment suggesting that bacteria can transfer genetic information through a process known as transformation.

Griffith's findings were followed by research that isolated DNA as the material that communicated this genetic information.

Q : What is struggle for survival ?

A : Darwin noted it. It is increade varieity. Every being must suffer destruction during some period of its life.

Q : What is the principles of natural selection ?

A : - Populations grow exponentially
- Variation occurs within species
- Traits can be inherited

Q : Why do you think Morgon used Drosophila in his studies. ?

A : It have short life cycle. Cost is low. It can produce huge number of offspring.

Q : Briefly explain Protein Structures.

A : Primary – Exact sequence of amino acids before folding
Secondary – Simple folding create simple structures
Tertiary – Folding results in complex 3D structures
Quaternary – Multiple 3D subunits orginazed into a bigger structure

Q : What do you expect for the Future of Science and Technology? User your own words .

A : People may find a treatment for cancer and other diseases, maybe we can avoid to be hungry eating pill. As everybody wants, robots are developed and they can do what we need.

Q : Explain the relationship between biochemistry, molecular biology and genetics

A : - Biochemistry focused on the protein part of life functions. It studies the components independent of the organism.

- Genetics focuses on the gene part. Usually mutants are used. So, it is organism without the component.

- Molecular Biology integrates those two, as can be quite well ascertained from the "central dogma".

Q : Briefly explain cell theory with respect to plant and animal cells.

A : The Cell Theory is - All living organisms are composed of cells. They may be unicellular or multicellular.
- The cell is the basic unit of life.
- Cells arise from pre-existing cells.

Larger organisms, such as a dog or a tomato plant, are also made up of cells. You could take a small tissue sample and place it under a microscope to see one actual cell within the larger living thing.

Q : Explain the work of Louis Pasteur and its significance.

A : Louis Pasteur's pasteurization experiment illustrates the fact that the spoilage of liquid was caused by particles in the air rather than the air itself. These experiments were important pieces of evidence supporting the idea of Germ Theory of Disease.

Q :Explain Mendels work and comment on his due dillegence.

A : Mendel was the first biologist to work out a detailed application at maths to biology. Mendel work about genetics. He grow and studied 29.000 pea plants. For his experiments, he selected the common pea plant and in preliminary testing identified 7 pairs of characteristic for study (form, color, length, etc.)

Q : What are the landmark achievements that led the development of Molecular Biology?

A : - The existence of DNA
- The elucidation of the structure of DNA
- The understanding how DNA controls the structure of the proteins made by cells.

Q : At what AGE do you think we are living now and why ?

A : I think we are living in technological age because there are many technological items which are developed everyday. And also people don't have to memorize many different information they can find them on the Internet.